College of Technology Galle

Course: Full Stack Software Development — Object Oriented Programming — Module K72T004M03 — NVQ Level 05

SEMESTER MODEL PAPER 04

Focus: Encapsulation practice, overloading intro, memory basics, inheritance vocab

Difficulty: Easy-Moderate

Duration: 3 Hours **Total Marks:** 100 **Questions:** 7 structured questions (answer any **5**) **Each question:** 20 marks

Instructions to Candidates

1. Answer any **five (5)** questions out of the seven provided.
2. Each question carries **20 marks**. Maximum mark for this paper is **100**.
3. Write answers in the answer booklet. Do **not** write answers on this question paper.
4. Where Java code is required, write clear, syntactically correct Java as taught in this module (VS Code / JDK).
5. Show workings / short explanations where asked. Manage your time (about 35 minutes per question).
6. Mobile phones and programmable devices are not allowed.

Question 1 [20 Marks]

Design rule: hide data.

a. State the usual rule for fields vs methods regarding `public`/`private`. **(4 marks)**

b. Write an encapsulated `Employee` with private `name` and `salary`, getters, and a setter for salary that rejects values below 0. **(10 marks)**

c. Show illegal vs legal access from `main` (one illegal line commented with the reason). **(6 marks)**

Question 2 [20 Marks]

Method overloading (same name, different parameters).

a. What is method overloading? **(4 marks)**

b. Write three overloaded `print` methods: one for `int`, one for `String`, one for `int` and `String`. **(9 marks)**

c. How does Java decide which overloaded method to call? **(4 marks)**

d. Can two methods overload if they differ only by return type? Explain. **(3 marks)**

Question 3 [20 Marks]

Constructor overloading.

a. Write class `Member` with fields `id` and `name`, a no-arg constructor (default values), and a two-arg constructor. **(10 marks)**

b. In `main`, create one object with each constructor and print their data. **(6 marks)**

c. What does Java provide if you write no constructor at all? **(4 marks)**

Question 4 [20 Marks]

Stack vs heap (simple memory).

a. In simple words, what lives on the stack and what lives on the heap? **(6 marks)**

b. Explain why `Book a = b;` can make two variables affect the same book. **(8 marks)**

c. What does `null` mean? What exception appears if you call a method on `null`? **(6 marks)**

Question 5 [20 Marks]

Inheritance vocabulary.

a. Define inheritance and the is-a relationship with one example (e.g. Car / Vehicle). **(6 marks)**

b. Which keyword creates an inheritance link in Java? Write one class header example. **(4 marks)**

c. Who is the superclass and who is the subclass in `class Dog extends Animal`? **(4 marks)**

d. Can a subclass use private fields of the superclass directly? How should it access them? **(6 marks)**

Question 6 [20 Marks]

Variable scope (simplified).

a. Define scope in one sentence. **(3 marks)**

b. Classify `count` (field), `value` (parameter), and `next` (declared inside a method) by scope. **(6 marks)**

c. Why does a variable declared inside an `if` block fail to compile if used after the block? **(5 marks)**

d. Write a tiny example that needs `this.x = x;` and explain why. **(6 marks)**

Question 7 [20 Marks]

```
class Counter {
    private int n;
    public void bump() { n = n + 1; }
    public int getN() { return n; }
}
// main:
Counter c = new Counter();
c.bump(); c.bump();
System.out.println(c.getN());
```

Short code reading.

a. What is printed? Explain step by step. **(6 marks)**

b. Is `n` a field or a local variable? How do you know? **(4 marks)**

c. Add a constructor that sets `n` to a starting value supplied by the caller. **(6 marks)**

d. What happens if `main` tries `c.n = 100;`? **(4 marks)**

******* END OF QUESTION PAPER *******